Physics for Engineers 2 Lab

COEFFICIENT OF LINEAR EXPANSION

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OBJECTIVES

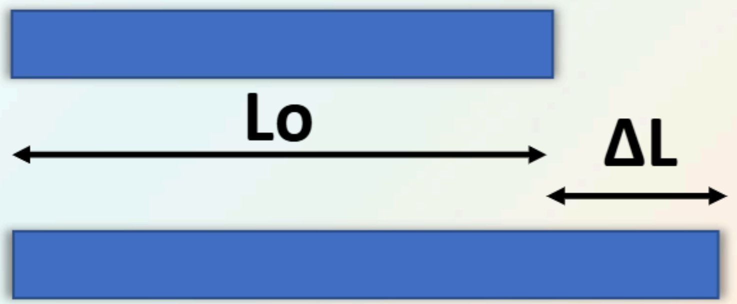
1. To measure the linear thermal expansion of some metal alloys.
2. To determine the linear expansion coefficients of some metal alloys.

Thermal Expansion

- Materials expand or contract when subjected to a change in temperature and the linear thermal expansion is, ΔL.

- The linear expansion is directly proportional to the object's length. The constant of proportionality is α, the coefficient of linear expansion of the material.

- Typically $\alpha \approx 10^{-5}/°C$ for metals



Coefficient of Linear Expansion

- In the experiment, you are to determine the coefficient of linear expansion of certain materials using the equation below

$$\alpha = \frac{\Delta L}{L_o \Delta T}$$

Note: This is the experimental value of the coefficient of linear expansion

Where

α is the coefficient of linear expansion (/°C)
ΔL is the change in length
L_o is the initial length
ΔT is the change in temperature
$\Delta T = T_f - T_i$

Percent Error

- The theoretical coefficient of linear expansion of steel and brass

$$\alpha_{steel} = 1.2x10^{-5}/C^{0}$$
$$\alpha_{brass} = 2.0x10^{-5}/C^{0}$$

- Compare the experimental coefficient of linear expansion with the theoretical by getting the percent error

$$\%\ error = \frac{|\alpha_{theo} - \alpha_{exp}|}{\alpha_{theo}} x100$$

DATA AND RESULT

The following tables are to be used in gathering data from the experiment. The data will be used in Summative Assessment.

Table 2.1 Linear Expansion of Brass			
Theoretical value of the coefficient of linear expansion (per °C) = 2×10^{-5}			
Description	**Initial Reading**	**Final Reading**	**Difference in Readings**
Length of the Metal Rod, mm			
Temperature of the Metal Rod. °C			

Table 2.2 Linear Expansion of Steel			
Theoretical value of the coefficient of linear expansion (per °C) = 1.2×10^{-5}			
Description	**Initial Reading**	**Final Reading**	**Difference in Readings**
Length of the Metal Rod, mm			
Temperature of the Metal Rod. °C			

Table 2.3 Percent Error of Coefficient of Linear Expansion		
	Brass	**Steel**
Percentage error between the experimental value and the theoretical value, % Error		

REFERENCES

Young, Hugh D. and Freedman, Roger A. (2010). *Sear's and Zemansky's University Physics* 12th Edition. United States of America: Addison Wesley Longman, Inc.

Giancoli, Douglas C. (2001). *Physics* 5th Edition. Pearson Education, Asia Pte Ltd.